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Removal of *Listeria monocytogenes* dual-species biofilms using combined enzyme-benzalkonium chloride treatments

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ABSTRACT

The effects of pronase (PRN), cellulase (CEL) or DNaseI alone or combined with benzalkonium chloride (BAC) against *Listeria monocytogenes*-carrying biofilms were assayed. The best removal activity against *L. monocytogenes*–*Escherichia coli* biofilms was obtained using DNaseI followed by PRN and CEL. Subsequently, a modified logistic model was used to quantify the combined effects of PRN or DNaseI with BAC. A better BAC performance after PRN compared to DNaseI eradicating *L. monocytogenes* was observed. In *E. coli* the effects were the opposite. Finally, effects of DNaseI and DNaseI–BAC treatments were compared against two different *L. monocytogenes*-carrying biofilms. DNaseI–BAC was more effective against *L. monocytogenes* when co-cultured with *E. coli*. Nonetheless, comparing the removal effects after BAC addition, these were higher in mixed-biofilms with *Pseudomonas fluorescens*. However, a high number of released viable cells was observed after combined treatments. These results open new perspectives of enzymes as an anti-biofilm strategy for environmental pathogen control.

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monocytogenes*

Introduction

In recent decades biofilms have been considered a major issue of concern in food processing related environments due to their relevance as a source of product contamination that results in product loss and contributes to food related illnesses caused by contaminated foodstuffs (Brooks & Flint 2008; Simões et al. 2010).

Listeria monocytogenes is a Gram-positive foodborne pathogen known to live as a saprophyte in environments rich in decaying plant material. This bacterium can cause listeriosis, an invasive disease that primarily affects the elderly, newborns, pregnant women and immunocompromised individuals (Farber & Peterkin 1991; Swaminathan & Gerner-Smidt 2007), with symptoms that may include sepsis, meningitis and miscarriages (Vázquez-Boland et al. 2001; Freitag et al. 2009). The European Food Safety Authority reported an incidence of 0.52 cases per 100,000 inhabitants of confirmed European *L. monocytogenes* infections, 30% higher regarding previous published data (EFSA 2015). Moreover, this microorganism is well known to survive for long periods attached to food industry surfaces (Borucki et al. 2003; Carpentier & Cerf 2011) as part of multi-species sessile communities

(Carpentier & Chassaing 2004; Rodríguez-López et al. 2015).

Chemical disinfectants such as acids, peroxides, sodium hypochlorite and quaternary ammonium compounds (QACs) have been used extensively in industrial settings for disinfection and control of biofilms (da Silva & De Martinis 2013). Benzalkonium chloride (BAC) is considered one of the most used QACs due to its action upon bacterial membranes, altering their structural integrity (Gerba 2015). Nevertheless, it has been extensively demonstrated that biofilms exhibit higher resistance/tolerance to BAC compared to planktonic cells both in Gram-positives such as *L. monocytogenes* (Saá Ibusquiza et al. 2011; Nakamura et al. 2013) and in Gram-negatives such as *Escherichia coli* (Houari & Di Martino 2007) or *Pseudomonas* sp. (Giaouris et al. 2013).

In the last few years, enzymes have increasingly been used for biofilm control. These environmentally friendly compounds have been shown to both prevent the initial adhesion and remove formed structures (Johansen et al. 1997; Orgaz et al. 2006; Lequette et al. 2010; Cordeiro & Werner 2011) because of their dispersive effect on the sessile structures acting on target molecules present in the

biofilm matrix (Giaouris et al. 2014; Kaplan 2014; Bridier et al. 2015). However, enzymes do not necessarily have bactericidal activity, which makes them unsuitable to be used as a strategy for disinfection (Nguyen & Burrows 2014).

To overcome this, a feasible strategy to obtain both biofilm disinfection and removal would be the combination of an enzyme and BAC solution. Although enzyme-based cleaners and detergents have been proved to be effective for biofilm removal (Parkar et al. 2004; Vickery et al. 2004; Stiefel et al. 2016), to the best of the authors' knowledge, the only evidence found for an enzyme–BAC combination against biofilms was the study performed by Kaplan (2009) who demonstrated that 24 h *Staphylococcus aureus* biofilms pre-treated with DNaseI were more sensitive to BAC and achieved about a 4 log CFU reduction in the remaining adhered cells when compared to the non-pre-treated samples.

The main aim of the present work was to utilise an approach using both enzyme treatment and subsequent disinfection using BAC on dual-species biofilms containing *L. monocytogenes* in an effort to facilitate the effect of the chemical disinfectant. For this purpose, classical plate counts as well as epifluorescence microscopy were used to assess the effects of the application of different enzyme solutions (pronase, cellulase or DNaseI) on *L. monocytogenes*–*E. coli* fluorescent tagged biofilms grown on stainless steel coupons. Next, a comparative study to quantify the effectiveness of combining the enzymatic solutions with BAC against 48 h *L. monocytogenes*–*E. coli* biofilms was carried out. Finally, the more effective enzymatic solution against the elimination of *L. monocytogenes* from the mixed biofilms was applied alone and combined with BAC against *L. monocytogenes*–*E. coli* and *L. monocytogenes*–*Pseudomonas fluorescens* in order to quantify the influence of the species composition on the effectiveness of cleaning and disinfection method.

Methods

Bacterial strains

Listeria monocytogenes A1 and *Escherichia coli* A14 were isolated from a fish processing plant in a previous survey (Rodríguez-López et al. 2015). *Pseudomonas fluorescens*

B52, a strong biofilm former and associated with milk and dairy products spoilage, was kindly provided by Dr Carmen San José (Allison et al. 1998).

In all situations, stock cultures were kept at -80°C in brain–heart infusion broth (BHI, Biolife, Milan, Italy) containing 50% glycerol 1:1 (v v^{-1}) mixed. Work cultures were kept at -20°C in tripticase soy broth (TSB, Cultimed, Barcelona, Spain) containing 50% glycerol 1:1 (v v^{-1}) mixed.

Construction of fluorescent-tagged stains

Genetic modification for constitutive expression of a fluorescent reporter of strains *L. monocytogenes* A1 and *E. coli* A14 was carried out in the laboratory of Prof. Colin Hill (School of Microbiology, University College Cork (UCC), Cork, Ireland).

Modification of *L. monocytogenes*

L. monocytogenes was modified for green fluorescent protein (GFP) constitutive expression. Briefly, the fragment of pNF8 corresponding to the *Pdlt*Ω*gfp-mut1* (Fortineau et al. 2000) was amplified with primers *Pdlt* For-*KpnI* and GFP pNF Rev-*PstI* (Table 1) containing *KpnI* and *PstI* restriction sites, respectively, digested and cloned into pPL2 (Lauer et al. 2002) previously digested with *KpnI* and *PstI* and further treated with rAPid alkaline phosphatase (Roche, Mannheim, Germany) to avoid religation. Ligation was performed using T4-ligase (Roche) in a PCR thermocycler as follows: 4°C for 5 h, 12 h ramp increasing 1°C h^{-1} , 16°C for 2 h and back to 4°C giving a plasmid of 7,393 bp coded as pROLO1. The plasmid solution was dialysed in sterile deionised water on a 0.025 µm pore nitrocellulose filter (Millipore, Darmstadt, Germany) for 30 min and then kept at -20°C until use. pROLO1 was then introduced into *E. coli* TOP10 cells (Thermo Fisher Scientific, Waltham, MA, USA) according to the manufacturer's instructions and cultured overnight in LB (Merck, Darmstadt, Germany) together with $10\text{ }\mu\text{g ml}^{-1}$ chloramphenicol (Cm, Sigma-Aldrich, St Louis, MO, USA) at 37°C . Plasmid extraction was then performed using a Gene JET Plasmid MiniPrep Kit (Thermo Fisher Scientific). *Pdlt*Ω*gfp-mut1* integration was checked with primers pPL2 MCS-for and pPL2 MCS-rev (Table 1).

Table 1. Sequences of the primers used in this work.

Primer	Sequence (5'→3')	Reference
<i>Pdlt</i> For- <i>KpnI</i>	TGGGTACCATTATACTCGTACCTAC	This study
GFP pNF Rev- <i>PstI</i>	AAACTGCA ^{TT} TATTGTATAGTTTCATCCATGCCA	This study
MCS for	GACGTCAATACGACTCACTATAGG	This study
pPL2 MCS-rev	GATAATAAGCGGATGAATGGCAG	This study
PL95	ACATAATCAGTCCAAAGTAGATGC	Bron et al. 2006
PL102	TATCAGACCTAACCCAAACCTTCC	Bron et al. 2006

Note: For: forward; rev: reverse; underlined: restriction site.

Electroporation was carried out by mixing 50 µl of electrocompetent cells prepared as previously described (Monk et al. 2008) with 2 µl of plasmid prep in 2 mm cuvettes using a BTX ECM 630 Generator (Harvard Apparatus, Holliston, MA, USA). Conditions: field strength: 10 kV cm⁻¹; time constant: 5 ms; voltage: 2 kV; resistance: 400 Ω; capacitance: 25 µF. Cells were then resuspended in fresh sterile BHI + 0.5 M sucrose, incubated at 37°C for 1 h and then plated on BHI + 1.5% agar + 10 µg ml⁻¹ Cm and incubated at 37°C for 48 h. Colonies were picked and PCR was performed to check for plasmid integration using primers PL95 and PL102 (Bron et al. 2006) (Table 1). The resulting isolate was named *L. monocytogenes* A1-gfp.

Modification of *E. coli*

E. coli was modified for mCherry constitutive expression using the λ-red system (Serra-Moreno et al. 2006; Hillyar 2012). *E. coli* A14 electrocompetent cells prepared as previously described (Gonzales et al. 2013) using 10% glycerol for the final cell resuspension. Then they were transformed with the thermosensitive plasmid pKO-BEGA, analogue to pKOBEG (Chaveroche et al. 2000) in which the *cat* gene has been substituted by a *bla*_{amp} gene (Sutcliffe 1978). This plasmid also contains the genes *exo*, *bet* and *gam*, necessary for λ-red system-mediated recombination (Chaveroche et al. 2000). Electroporation was carried out in 2 mm cuvettes in a BTX ECM 630 generator. Conditions: field strength: 10 kV cm⁻¹; time constant: 5 ms; voltage: 2.5 kV; resistance: 200 Ω; capacitance: 25 µF. Transformants were selected on LB agar + 50 µg ml⁻¹ ampicillin (Amp, Sigma-Aldrich) at 30°C for 24 h.

Then, *E. coli* A14 pKOBEGA electrocompetent cells were prepared as above and newly transformed with pMP7607 miniTn7 (Lagendijk et al. 2010) carrying the mCherry gene and a streptomycin (Sm) resistance gene. Transformants were selected onto LB agar + 50 µg ml⁻¹ Amp + 50 µg ml⁻¹ Sm (Sigma-Aldrich) and incubated at 30°C for 24 h. Fifty randomly chosen transformants were picked and spread onto LB agar + 50 µg/ml Sm and incubated at 42°C. The resulting isolate was named *E. coli* A14-mChy.

To assess the correct fluorescent signal, ten randomly picked colonies of each modified strain were diluted in a drop of deionised water on a glass slide and visualised under the fluorescence microscope.

Biofilm set-up

One hundred microlitres of work cultures was grown overnight at 37°C in 5 ml of BHI + 10 µg ml⁻¹ Cm for *L. monocytogenes* A1-gfp and LB + 50 µg ml⁻¹ Sm for *E. coli* A14-mChy and subcultured overnight so as to ensure proper growth.

Inocula preparation was performed following a modification of a protocol previously described (Rodríguez-López et al. 2015). Briefly, cultures were adjusted to Abs₇₀₀ = 0.1 ± 0.001 in sterile phosphate buffer saline (PBS) using a Cecil3000 scanning spectrophotometer (Cecil Instruments, Cambridge, UK), corresponding to a concentration of about 10⁸ CFU ml⁻¹. Adjusted cultures were further diluted in sterile mTSB (TSB supplemented with 2.5 g l⁻¹ glucose (Vorquímica, S.L., Vigo, Spain) and 0.6% yeast extract (Cultimed, Barcelona, Spain)) to a final concentration of about 10⁴ CFU ml⁻¹. Then, equal volumes of these adjusted cultures were mixed to obtain the inoculum for dual-species biofilms.

Biofilms were grown on 10×10×1 mm AISI 316 stainless steel (SS) coupons (Comevisa, Vigo, Spain). Pre-treatment of coupons included individual washing with industrial soap (Sutter Wash, Sutter Ibérica, S.A., Madrid, Spain), rinsing with tap water, a final rinse with deionised water and autoclaving at 121°C for 20 min. Coupons were then placed individually into a 24 flat-bottomed well plate and each well was inoculated with 1 ml of the corresponding culture. Plates were incubated in a humidified atmosphere at 25°C statically for 2 h so as to allow initial adhesion, and then in constant shaking at 100 rpm.

Biofilm formation kinetics

Samples (SS coupons) were collected at 24, 36, 48, 72 and 100 h and briefly immersed in sterile PBS in order to remove loosely attached cells before any analysis was performed.

Determination of the number of adhered viable cells (AVC)

Three different coupons were scraped using two cotton swabs pre-moistened with buffered peptone water (BPW; Cultimed, Barcelona, Spain). The swabs were then placed in 2 ml of BPW vigorously vortexed for 1 min to resuspend the cells. The cell suspensions were then serially diluted in BPW and spread in duplicates onto agar plates. *Listeria*-PALCAM (Liofilchem, Roseto degli Abruzzi, Italy) was used to select *L. monocytogenes* and HiCrome™ Coliform agar (Sigma-Aldrich) with a supplement of 5 µg ml⁻¹ of vancomycin and cefsulodine (Sigma-Aldrich) for *E. coli* selection. Plates were incubated at 37°C for 24–48 h and results were expressed as the mean in log CFU cm⁻² of samples. The accepted limit of detection for this and all assays involving viable cell counts was at least 25 CFU in the plate of the lowest dilution corresponding to a total of 1.70 log CFU cm⁻² (Sutton 2011).

Epifluorescence microscope visualisation

At each sampling time, three coupons were air dried avoiding as much as possible direct light exposure. Samples

were then visualised under a Leica DM6000 epifluorescence microscope (Leica, Wetzlar, Germany) using a 40× objective and 10× ocular lenses. Microscope was equipped with filter cubes L5 (Excitation 480/40) for A1-gfp and TX2 (Excitation 560/40) for A14-mChy. Images were taken using a Leica DFC365 FX controlled with Metamorph MMAF software (Molecular Devices, Sunnyvale, CA, USA, camera from 10 representative fields).

Effect of enzymatic solutions on dual-species biofilms

Enzyme solutions were prepared at concentrations 200, 400, 700 and 1,000 µg ml⁻¹. Pronase (PRN, from *Streptomyces griseus*, Roche) was dissolved in 0.1 M Tris-HCl (Sigma-Aldrich) buffer at pH 7.5 ± 0.2. Cellulase (CEL, from *Aspergillus niger*, Sigma-Aldrich) was dissolved in 100 mM citrate (Sigma-Aldrich) buffer at pH 6.0 ± 0.1. DNaseI (from bovine pancreas, Sigma-Aldrich) was dissolved in 10 mM Tris-HCl (pH 7.5 ± 0.2) buffer also containing 2.5 mM MgCl₂ and 0.1 mM CaCl₂. After preparation, all solutions were filter sterilised through a 0.2 µm syringe filter (Sartorius, Göttingen, Germany) and kept at -20°C until use.

The biofilm removal action of each enzymatic solution was evaluated against 24 h biofilms. Three coupons were washed as before and then placed in a clean well. One millilitre of each enzyme solution was added and allowed to act for 30 min at 37°C for PRN and 32°C for CEL and DNaseI. Negative controls were run in parallel by adding the corresponding buffer solution without enzyme. Solutions were then gently removed by pipetting and SS coupons were subsequently washed with 1 ml of sterile PBS in order to remove residual enzyme. Determination of remaining adhered cells and visualisation of coupons was performed as described above. Results were expressed as the reduction in log CFU cm⁻², calculated as the mean of each replica difference in log CFU cm⁻² before enzymatic and after enzymatic treatment. After this, the two most effective enzymes were used in the following experiments.

Effect of benzalkonium chloride combined with either PRN or DNaseI on *L. monocytogenes*-*E. coli* biofilms

Benzalkonium chloride solutions (BAC, Guinama, Alboraya, Spain) were prepared in sterile deionised water at concentrations 25, 50, 100, 250 and 500 µg ml⁻¹. Each solution was applied after 30 min treatment with 400 µg ml⁻¹ of either PRN or DNaseI solution against 48 h *L. monocytogenes* A1-gfp-*E. coli* A14-mChy biofilms.

Fourteen different coupons washed with sterile PBS for removal of loosely attached cells were used for each enzyme series: two for the negative controls (no

treatment), two for enzyme treatment without BAC (only enzyme and deionised water were applied), and two for each BAC concentration after enzymatic treatment performed as described above. In this latter case, 1.5 ml of each BAC solution were added to each coupon for a 10 min contact time at room temperature. For negative controls, buffer without enzymes and deionised water were sequentially used. Coupons were then transferred to a new well and immersed for 30 s in 1 ml of a neutralising solution (composition l⁻¹: 10 ml of a 34 g l⁻¹ KH₂PO₄ buffer (pH 7.2); soybean lecithin: 3 g; Tween 80: 30 ml; Na₂S₂O₃: 5 g; L-histidine: 1 g) at room temperature followed by a final 10 s wash by immersion with sterile PBS to remove any neutraliser residues.

Following its application, neutralising solution was serially diluted in BPW and spread in duplicate onto appropriate agar medium to determine the number of released viable cells (RVC) after treatments. Outcomes were expressed as mean of log CFU ml⁻¹. Microscopic visualisation and determination of the remaining attached cells were performed as described above. In the latter case, results were expressed as percentage of biofilm removal with respect to the log CFU cm⁻² obtained in the control samples.

Determination of the BAC effect: calculation of lethal dose 90 (LD₉₀)

LD₉₀, defined as the dose of an antimicrobial required to achieve a 90% kill of the initial bacterial population, was used as a parameter to determine the effect of BAC on dual-species biofilms. To assess this, a modified logistic model proposed by Cabo et al. (1999) was used. Logistic equations are widely recognised as suitable for describing dose-response kinetics (Knight & McKellar 2007; Murado & Vázquez 2007). Firstly, outcomes were obtained by fitting of the experimental data obtained in plate count assays, expressed in percentage of biofilm removal according to Equation 1 using the least-squares method (quasi-Newton) of the SOLVER tool of Microsoft Excel 2016:

$$BR = K \left(\frac{1}{1 + 0.11e^{r(LD_{90} - D)}} - \frac{1}{1 + 0.11e^{rLD_{90}}} \right) \quad (1)$$

where BR = biofilm removal expressed as a percentage; LD₉₀ = the dose of BAC that removes 90% of the initial adhered population; D = dose of BAC used; K = the maximum percentage of biofilm removal (asymptote); and r = the specific inhibition coefficient (dimensions: inverse of the dose).

Since Equation 1 modifies the resulting dose/response parameters by subtracting the intercept of the original logistic equation, the results were further adjusted to obtain the new K value (K’):

$$BR_{\max} = K' = \lim_{D \rightarrow \infty} BR \quad (2)$$

Then, the real LD_{90} (RD_{90}) was determined according to a modification of an equation described previously (Murado et al. 2002):

$$RD_{90} = \frac{1}{r} \ln(9 + e^{rD}) \quad (3)$$

Influence of *L. monocytogenes* accompanying species in the resistance to DNaseI–BAC treatments

Two different 48 h dual-species biofilms were used: *L. monocytogenes* A1–*E. coli* A14 and *L. monocytogenes* A1–*P. fluorescens* B52 to evaluate sequential DNaseI–BAC treatments.

400 $\mu\text{g ml}^{-1}$ of DNaseI + 100 $\mu\text{g ml}^{-1}$ of BAC treatment and plate count analysis for attached and released cells determination were performed as described above. For *P. fluorescens* selection *Pseudomonas* agar base (PAB, Liofilchem) supplemented with CFC supplement (Liofilchem) was used and incubated at 30°C for 48 h.

For microscopic visualisation, samples were stained using LIVE/DEAD bacterial viability kit (Life Technologies, Eugene, OR, USA) to distinguish total cells with undamaged membranes (green fluorescence) and damaged cells (red fluorescence). Staining solution was prepared by mixing 0.25 μl of propidium iodide and 0.75 μl of Syto9 in 1 ml of filter sterilised deionised water. Fifty microlitres of this solution were then poured onto each coupon and allowed to remain for 15 min in the dark. Coupons were then washed three times in 1 ml of sterile MilliQ water, air dried and visualised under an epifluorescence microscope to obtain images of representative fields.

Statistical analysis

The experimental results were analysed for statistical significance using IBM SPSS Statistics for Windows, Version 23.0 (IBM Corp., Armonk, NY, USA). An independent-samples two-tailed Student's *t* test was performed to assess differences between species in the biofilm formation kinetics and the effects of BAC in RVC after PRN and DNaseI treatments. Differences among the effects of the different enzymatic treatments and treatments' effects in different dual-species biofilms were determined using a one-way ANOVA with a *post hoc* Bonferroni test. In all cases, significance was expressed at the 95% confidence level ($\alpha = 0.05$) or greater.

In RD_{90} determinations, the correlation coefficient (r^2) was calculated to quantify the discrepancy between the observed experimental values and those expected according with the model.

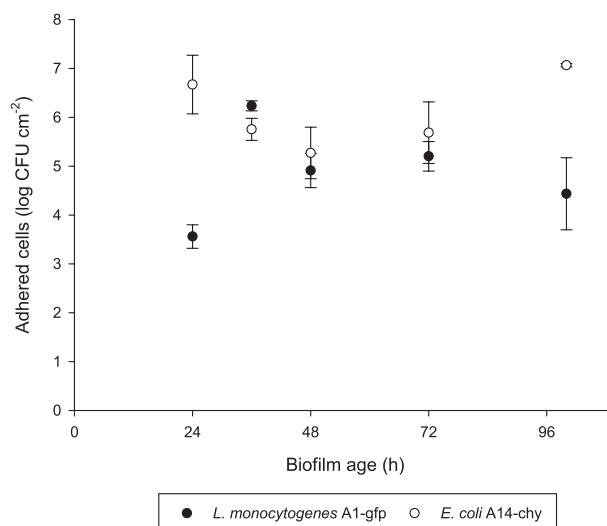


Figure 1. Growth dynamics of the *L. monocytogenes*-gfp–*E. coli*-mChy dual-species biofilm. Asterisks indicate statistically significant differences ($\alpha = 0.05$). Error bars = SD values of each sampling time dataset ($n = 3$).

Results

L. monocytogenes–*E. coli* biofilm formation kinetics on AISI 316 SS

Dual-species biofilm formation dynamics are depicted in Figure 1. Plate count assays showed a significantly higher number of AVC in *E. coli* with respect to that obtained for *L. monocytogenes* at 24 and 100 h of growth, yielding differences of 3.11 and 2.63 log CFU cm^{-2} respectively. No significance was observed among the values of the rest sampling times.

Microscopic images displayed in Figure 2 showed a uniform distribution of *E. coli* and *L. monocytogenes* over the coupon. In spite of this uniform distribution, at 24 h *E. coli* presented ~3 log higher AVC counts compared to *L. monocytogenes* (Figure 1). A tendency for aggregation was observed at 24 and 36 h yielding a final composite structure with both species intermingled therein (Figure 2). From this point onward, the amount of cells increased and the biofilm developed a cloud-shape structure which was maintained in the last three sampling times (Figure 2).

Effects of pronase, cellulase and DNaseI on the elimination of mixed biofilms formed by *L. monocytogenes*–*E. coli*

The effects of the application of PRN, CEL and DNaseI on the number of AVC of 24 h *L. monocytogenes*–*E. coli* biofilm were compared. The results are expressed in terms of log CFU cm^{-2} reduction (Figure 3).

In general terms, *L. monocytogenes* was more sensitive than *E. coli* to the treatments used, yielding higher log

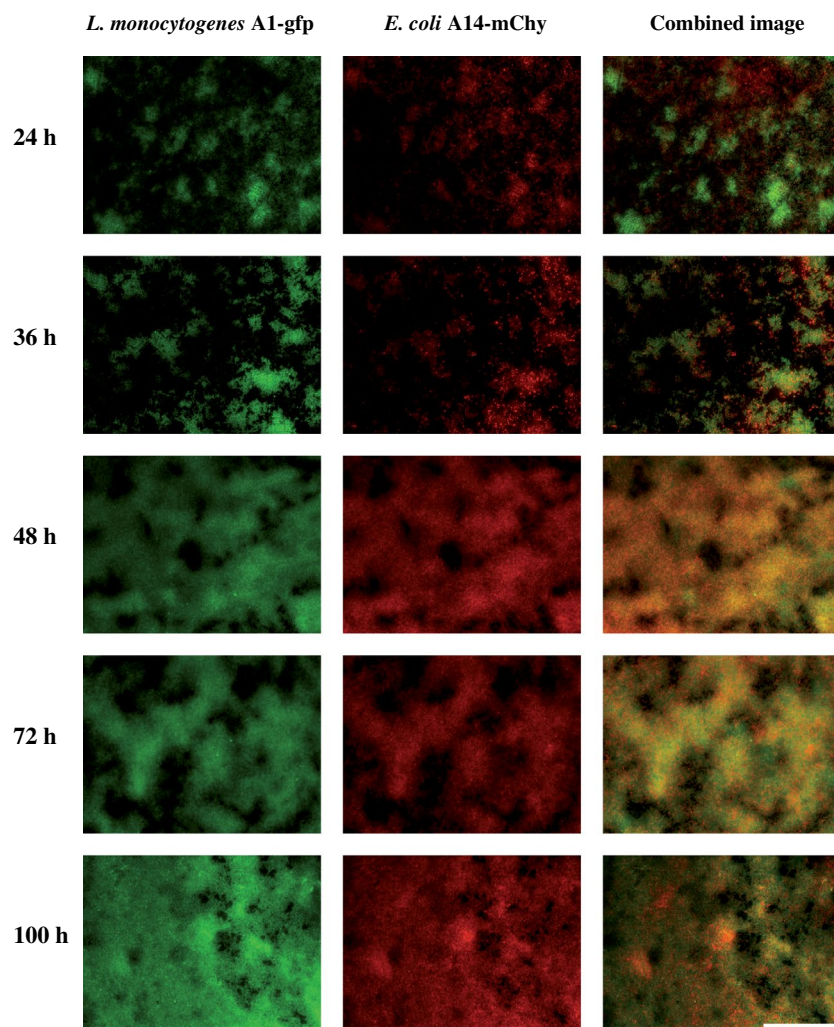


Figure 2. Fluorescence microscopy 40 $\times$ -field images of *L. monocytogenes* A1-gfp, *E. coli* A14-mChy and combined fields in dual-species biofilm formation kinetics. Scale bar = 100 μ m.

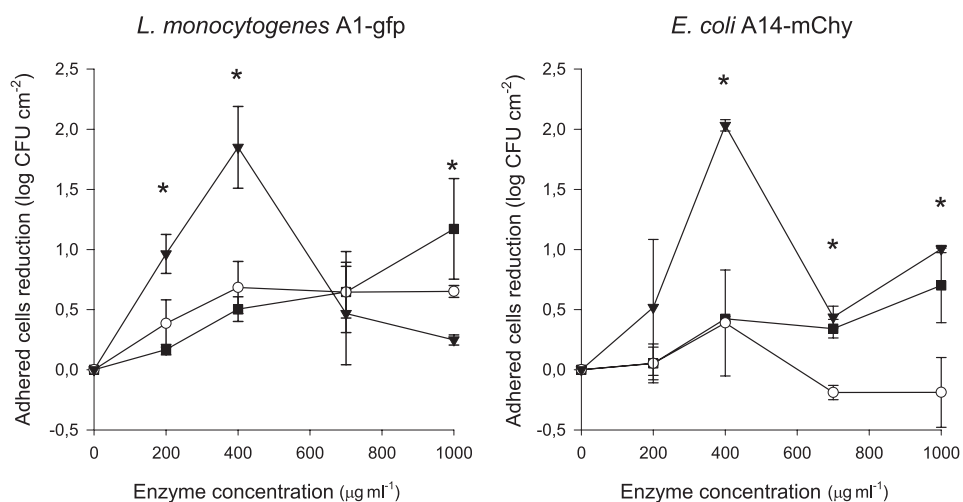


Figure 3. Logarithmic reductions of adhered cells obtained on 24 h *L. monocytogenes* A1-gfp–*E. coli* A14-mChy dual-species biofilms after an enzymatic treatment with PRN (■), CEL (○) or DNaseI (▼). Asterisks indicate statistically significant differences in any of the treatments at a given concentration. Error bars = SD of each dataset ($n = 3$).

reductions in most of the concentrations and enzymes used, with the exception of DNaseI at 1,000 $\mu\text{g ml}^{-1}$, where *E. coli* log reductions were significantly higher (Figure 3). Comparing the effects of the enzymes, higher concentrations were required to achieve a comparable log reduction of AVC in *E. coli* being especially relevant in the case of PRN and CEL (Figure 3).

In both species, maximum effects (about a 2 log reduction) were obtained after the application of 400 $\mu\text{g ml}^{-1}$ of DNaseI. In *L. monocytogenes*, the log reduction value was significantly higher when treated with DNaseI compared to treatments with PRN and CEL in two out of four concentrations tested (200 and 400 $\mu\text{g ml}^{-1}$) (Figure 3). On the other hand, considering *E. coli* removal by DNaseI, significance was only observed after applying a 400 $\mu\text{g ml}^{-1}$ of solution (Figure 3). In both species, application of higher concentrations of this enzyme resulted in a lower log reduction. Indeed, biofilm removal decreased about 1.5 log CFU cm^{-2} when the DNaseI concentration applied was increased from 400 to 600 $\mu\text{g ml}^{-1}$.

The application of CEL resulted in lower log reductions in both species tested compared to outcomes obtained after treatment with DNaseI, with the exception of 1,000 $\mu\text{g ml}^{-1}$ against *L. monocytogenes* where CEL significantly performed better than DNaseI (Figure 3).

Finally, the results displayed a concentration-dependent increase in log reduction in both species when PRN was used, with maximum log reductions at 1,000 $\mu\text{g ml}^{-1}$ of 1.17 ± 0.42 and 0.70 ± 0.31 log CFU cm^{-2} for *L. monocytogenes* and *E. coli*, respectively (Figure 3).

Combined effects of BAC and PRN or DNaseI solutions for 48 h *L. monocytogenes*–*E. coli* biofilm elimination

The maximum percentage of biofilm removal (K') and the lethal dose 90 (RD_{90}) values for *L. monocytogenes* and *E. coli* were calculated according to Equations 1–3 after sequential treatment with 400 $\mu\text{g ml}^{-1}$ of either PRN or DNaseI followed by disinfection with different concentrations of BAC. In these experiments 48 h biofilms were preferred to provide a more challenging scenario to enzyme–BAC treatments.

The results showed a satisfactory fitting of experimental data ($r^2=0.984$) and demonstrated a higher efficacy of both combined treatments removing *L. monocytogenes* with respect to *E. coli* as indicated by K' values (Figure 4, Table 2). Whereas in the case of *L. monocytogenes* BAC performed better after DNaseI treatment compared to PRN, in *E. coli* RD_{90} values showed a higher effect of BAC after PRN treatment compared to DNaseI (Table 2).

The outcomes of RVC (*L. monocytogenes* and *E. coli*) demonstrated a high level of cell dispersion after the

application of sequential enzyme–BAC treatments, with values ranging from ~ 3 –5 log CFU ml^{-1} (Figure 5). The Student's t test showed significance ($p < 0.05$) between treatments at BAC concentrations of 25 and 50 $\mu\text{g ml}^{-1}$ in *L. monocytogenes* and 25 and 100 $\mu\text{g ml}^{-1}$ in *E. coli*, with a general tendency to lower RVC values as the BAC concentration increased (Figure 5). If only RVC values of *L. monocytogenes* are considered, is important to highlight that no viable cells were detected after ≥ 100 $\mu\text{g ml}^{-1}$ BAC either in PRN or in DNaseI-treated samples (Figure 5).

Role of the accompanying species (*E. coli*, *P. fluorescens*) in the adhesion and resistance of *L. monocytogenes* to DNaseI and DNaseI–BAC treatments in dual-species biofilms

Cell counts demonstrated that *L. monocytogenes* was able to achieve a significantly higher number of adhered cells in the presence of *P. fluorescens* compared to co-culture with *E. coli*, reaching values of 7.23 ± 0.04 and 5.48 ± 0.05 log CFU cm^{-2} , respectively (Figure 6(A) and (B)).

The application of a 400 $\mu\text{g ml}^{-1}$ of DNaseI solution gave higher *L. monocytogenes* log reduction values in co-culture with *E. coli* (2.47 log CFU cm^{-2}) compared to that obtained in co-culture with *P. fluorescens* (0.58 log CFU cm^{-2}) (Figure 6(A) and (B)). Combined treatments (400 $\mu\text{g ml}^{-1}$ DNaseI + by 100 $\mu\text{g ml}^{-1}$ BAC) also produced a significant reduction in *L. monocytogenes* compared to the controls, of 3.24 and 2.83 log CFU cm^{-2} in co-culture with *E. coli* and *P. fluorescens*, respectively (Figure 6(A) and (B)). Nevertheless, if only the BAC effects on *L. monocytogenes* are considered, by comparing the log reductions of DNaseI alone and DNaseI–BAC treatments, these were higher in *L. monocytogenes*–*P. fluorescens* biofilms (2.55 log CFU cm^{-2}) compared to *L. monocytogenes*–*E. coli* samples (0.77 log CFU cm^{-2}) (Figure 6(A) and (B)).

L. monocytogenes RVC after DNaseI–BAC treatment did not present significant differences comparing both dual-species biofilms (4.23 ± 0.41 log CFU ml^{-1} in *L. monocytogenes*–*P. fluorescens* and 3.65 ± 0.41 log CFU ml^{-1} in *L. monocytogenes*–*E. coli*). Note that *E. coli* presented a significantly higher number of RVC (6.22 ± 0.09 log CFU ml^{-1}) after a DNaseI–BAC combined treatment compared with *P. fluorescens* and *L. monocytogenes* in both dual-species biofilms (Figure 6(C) and (D)).

Microscopic analysis showed that both biofilms presented marked differences in their 2D morphologies (Figure 7). While *L. monocytogenes*–*E. coli* biofilms showed a reticular distribution in all biofilm, *L. monocytogenes*–*P. fluorescens* biofilms were characterised by the presence of microcolonies surrounded by small cell groups. These microcolonies presented a local accumulation of damaged cells compared to the rest of the sample,

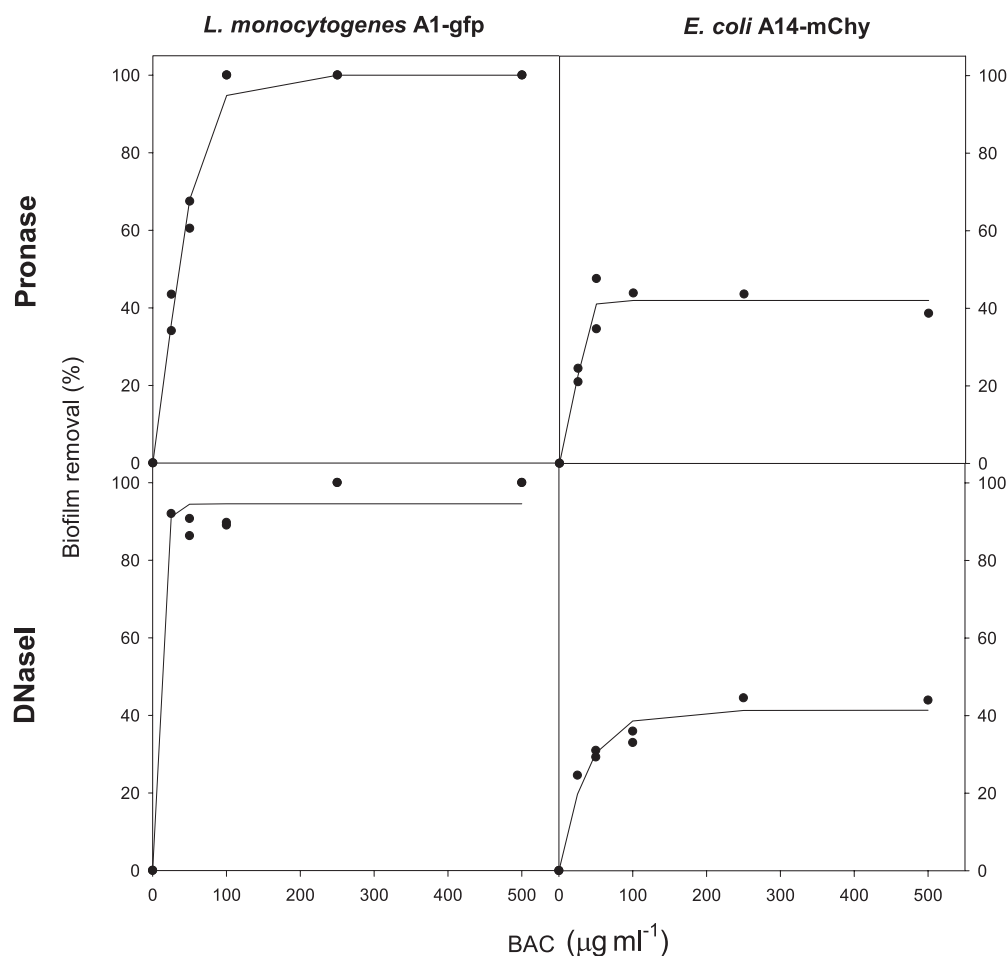


Figure 4. Lethal dose 90. Fit of biofilm removal values against *L. monocytogenes*–*E. coli* mixed biofilms obtained after the application of PRN–BAC or DNaseI–BAC treatments according to Equation 1.

Table 2. Parameters obtained after fitting biofilm removal experimental data to Equations 1–3.

	<i>L. monocytogenes</i> A1-gfp		<i>E. coli</i> A14-chy	
	K' (%)	BAC RD ₉₀ (mg kg ⁻¹)	K' (%)	BAC RD ₉₀ (mg kg ⁻¹)
Pronase	100.00	82.28	42.06	38.90
DNaseI	94.59	16.74	41.39	82.10

Notes: Maximum percentage reduction (K') and real lethal dose 90 (RD₉₀) values obtained due to BAC action after a single application of a 400 μg ml⁻¹ solution of either pronase or DNaseI.

as observed by a higher red signal in the central part of the microcolony. The same microcolony formation tendency was also observed in the authors' laboratory with other *L. monocytogenes* strains when co-cultured with *P. fluorescens* B52 (data not shown).

Sequential DNaseI–BAC treatments produced a significant increase in the red cell signal especially in *L. monocytogenes*–*E. coli* samples, indicating the BAC killing effects. Besides, noticeable structural changes were observed in samples of both dual-species biofilms, especially in *L. monocytogenes*–*P. fluorescens* biofilms in

which the cellular groups surrounding the microcolonies were substituted by sparsely distributed cells (Figure 7).

Discussion

The biofilm kinetics of the *L. monocytogenes* A1-gfp–*E. coli* A14-chy biofilm showed a typical biofilm fit-curve with minor fluctuations (Figure 1). Viable counts of *E. coli* were significantly higher than *L. monocytogenes* at 24 and 100 h. Differences in AVC counts at 24 h could be attributed to a better initial adhesion of *E. coli* compared with *L. monocytogenes* as previously reported (Azevedo et al. 2014). However, AVC values of both species were equilibrated at 36, 48 and 72 h (Figure 1).

Microscopic images showed a uniform distribution of *E. coli* and *L. monocytogenes* over the coupon despite the differences up to 3 log present between these species at 24 h (Figures 1 and 2). Almeida et al. (2011) reported that in 48 h *L. monocytogenes*–*E. coli* biofilms grown on SS and plastic, species presented this sort of uniform distribution with *E. coli* being present in higher numbers. The fact that green fluorescence, corresponding to *L. monocytogenes*

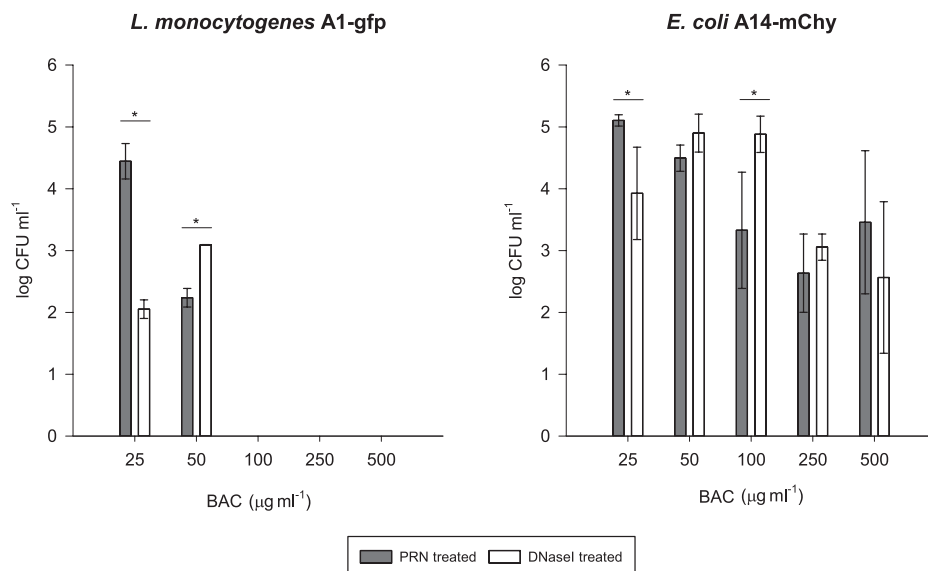


Figure 5. Released viable cells of *L. monocytogenes* (left) and *E. coli* (right) from 48 h dual-species biofilms after the application of different BAC solutions following a single dose of a 400 μg ml⁻¹ solution of pronase (filled bars) or DNaseI (void bars). Error bars = SD of each sample set. Asterisks indicate statistically significant differences between enzymatic treatments at each BAC concentration (α = 0.05).

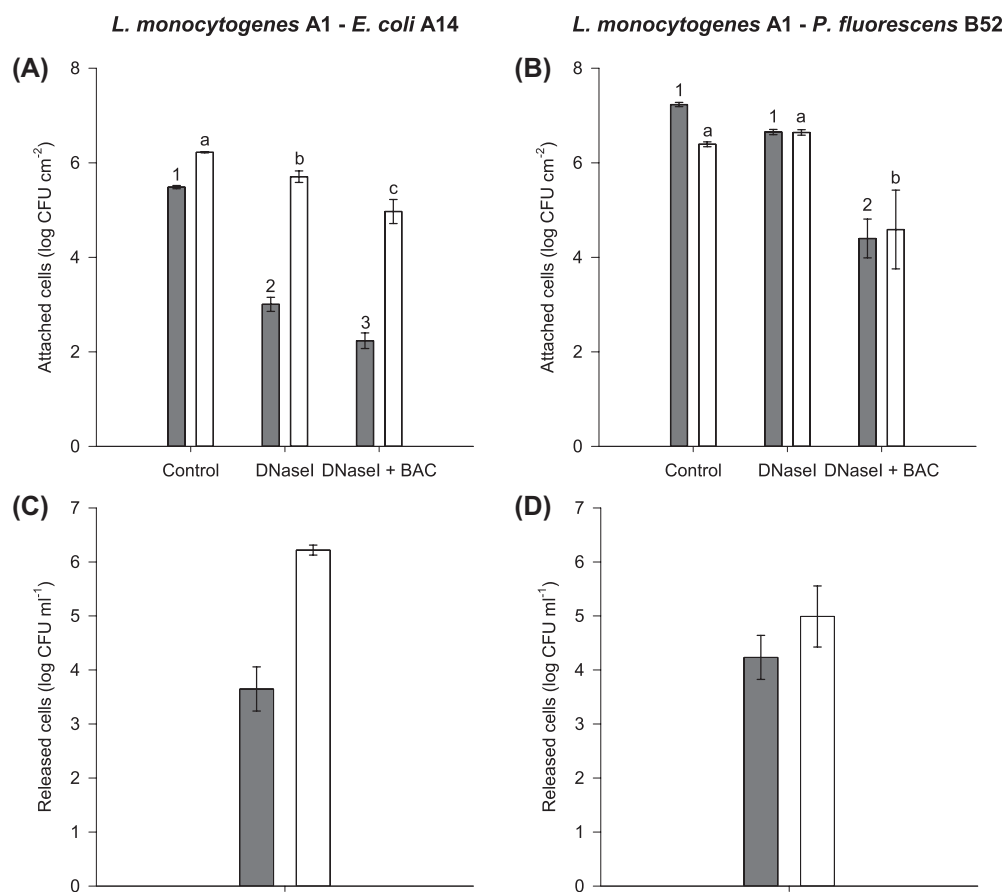


Figure 6. Sensitivity of 48 h *L. monocytogenes* A1 dual-species biofilms to the application of DNaseI and DNaseI-BAC. (A, B) Number of viable attached cells of *L. monocytogenes* (filled bars) and of *E. coli* A14 (A) and *P. fluorescens* B52 (B) (void bars). For each species separately, bars with different number or letter indicate significant differences (α = 0.05). (C, D) Number of viable released cells after the DNaseI-BAC treatment. Error bars represent the standard deviation of each sample set (n = 3).

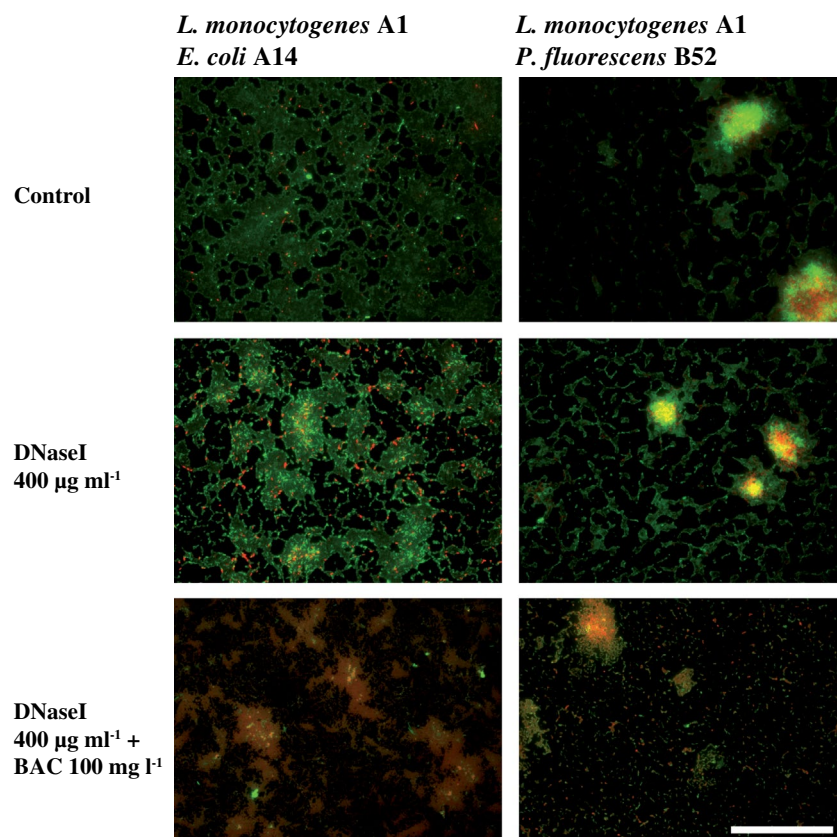


Figure 7. Fluorescence microscopy 40 $\times$ -field images for comparison of the effects of DNaseI–BAC combined treatments in two different 48 h *L. monocytogenes* dual-species biofilms. Scale bar = 100 μ m.

cells, was similar to red despite viable counts (Figure 1), could have been caused in part because a fraction of this green signal was emitted by cells in the viable but non culturable (VBNC) state. Previous authors have observed that 24 h-old *L. monocytogenes* biofilms present a part of VBNC (Gião & Keevil 2014). In such conditions, GFP remains totally functional and fluoresces even though cells are not able to grow in solid media (Cho & Kim 1999; Lowder et al. 2000).

Enzymes have been previously used as a biofilm removal strategy due to their specificity and their low environmental impact (Cordeiro & Werner 2011; Thallinger et al. 2013; Meireles et al. 2016). In this work, comparison between the effects of cellulase (CEL), DNaseI and pronase (PRN) demonstrated a maximum effect of a 400 μ g ml⁻¹ DNaseI solution reducing by about 2 log CFU cm⁻² the number of AVC in *L. monocytogenes* and *E. coli* (Figure 3). This reduction was followed by that produced by PRN and CEL, despite no broad differences being observed between these two (Figure 3).

It has been reported that extracellular DNA (eDNA) is present in considerable amounts of the extracellular matrix and is considered as a requisite for biofilm formation in *L. monocytogenes* (Harmsen et al. 2010) as well as in other Gram-positives (Qin et al. 2007; Vilain

et al. 2009). Hence, DNaseI has been proposed as an anti-biofilm enzyme cleaving eDNA and thus interfering in biofilm development. As an example, Harmsen et al. (2010) observed that a 100 μ g ml⁻¹ DNaseI solution at 37°C completely prevented *L. monocytogenes* EGDe biofilm formation if applied up to 24 h after strain inoculation and, from that point onwards, the DNaseI anti-biofilm capacity was reduced. In other Gram-positives such as *S. aureus*, a contact time of 1 h at 37°C of a 100 μ g ml⁻¹ DNaseI solution significantly reduced the biomass of 24 h biofilms grown on polystyrene plates (Izano et al. 2008). Despite these previously reported data, no complete removal with DNaseI was achieved among the experiments performed in this work. This could be due to the application of a more realistic time of action (30 min) or to the biofilm age, which could affect DNaseI biofilm removal activity (Harmsen et al. 2010). Experimental data also showed an inverted effect of DNaseI at concentrations higher than 400 μ g ml⁻¹ (ie higher doses produced a lower log reduction), both in *L. monocytogenes* and *E. coli* (Figure 3). Nguyen and Burrows (2014) demonstrated a similar enzymatic stimulatory effect on planktonic *L. monocytogenes* cells in which the more proteinase K present in the culture, the more stimulated its growth was. Focusing on the present

experimental approach, these effects in the number of cells in the planktonic state could have had eventually provoked an upturn in the number of cells adhered to the biofilm detected in AVC assays.

Proteases have also been proved to be effective in removing biofilms. In this context, Nguyen and Burrows (2014) demonstrated that the addition of 100 µg ml⁻¹ of proteinase K for 24 h was able to disperse 72 h *L. monocytogenes* biofilms grown on polystyrene up to undetectable levels. In *S. aureus* it has been recently reported that active proteases removed biofilms formed in polystyrene plates (Stiefel et al. 2016). However, PRN effects against *L. monocytogenes* were lower than expected compared with DNaseI, considering the proteinaceous nature of *L. monocytogenes* biofilm matrix (Combrouse et al. 2013; Nguyen & Burrows 2014) even though it has been demonstrated that teichoic acids are also present (Brauge et al. 2016).

Previous investigations have reported that interspecies interactions that take place within multi-species biofilms significantly modify the matrix composition if compared with monocultures (Giaouris et al. 2015; Sanchez-Vizueté et al. 2015). This differential composition can affect, among others, the efficacy of enzymes as well as several antimicrobial compounds (Sanchez-Vizueté et al. 2015). In the present case, the dominance of *E. coli* in 24 h biofilms (Figure 1) could have given rise to a matrix with a higher polysaccharide content, as proposed for most Gram-negative bacteria (Sutherland 2001). Nevertheless, CEL showed the lowest effects against *L. monocytogenes*–*E. coli* biofilms perhaps because the polysaccharide constituents interacted among themselves and among other molecules present, thus concealing enzyme targets or they simply lack glucose–glucose bonds susceptible to cleavage by CEL specific β (1→4) endoglucanase activity.

Considering the aforementioned results, it is logical to think that the use of dual-species biofilms represents a more challenging environment for biofilm-degrading enzymes due to a higher matrix complexity. Thus, the idea of a combination of enzymes would be an interesting option to be considered for proper biofilm removal (Simões et al. 2010; Meireles et al. 2016), especially when dealing with Gram-negatives such as *Pseudomonas* sp. (Stiefel et al. 2016). The efficacy of enzymatic mixtures has been previously reported by Orgaz et al. (2006) using proteinase, cellulase, pectinesterase, pectin lyase and alginate lyase derived from fungal cultures against 24-h-old *P. fluorescens* B52 biofilms on glass, achieving removal values up to an 84% of the total biomass.

In any case, enzymatic solutions show only a dispersing-but-not-killing effect as previously reported (Nguyen & Burrows 2014). As a consequence, enzyme based disinfection may need to be performed in combination with

biocides that are able to kill the cells avoiding the dispersion of live cells released from the biofilm (Meireles et al. 2016; Stiefel et al. 2016).

In food related premises, RVC could provoke a pathogen, thus enhancing the formation of new reservoirs and increasing the probability of product contamination. Also, pathogens could be easily spread through rinse after disinfection via water or aerosols produced (Todd et al. 2009) or by means of typical cleaning tools such as sponges or wipes (Kusumaningrum et al. 2003). Therefore, controlling RVC after cleaning and disinfection treatments appears to be as an interesting topic to consider for further investigation.

Enzyme–BAC combined treatments showed a differential effect on *L. monocytogenes*–*E. coli* biofilms depending on the species. More specifically, BAC performed better against *L. monocytogenes* when preceded by DNaseI, whereas removal of *E. coli* from the coupon was higher after PRN–BAC treatment (Table 2 and Figure 4). In *L. monocytogenes* the lower BAC RD₉₀ values obtained after DNaseI treatment indicated that despite proteins being considered the main fraction in *L. monocytogenes* biofilm matrix (Combrouse et al. 2013), eDNA degradation by DNaseI provokes a higher decrease in *L. monocytogenes* AVC counts, thus confirming the key role of eDNA in maintaining already formed biofilms (Harmsen et al. 2010; Nguyen & Burrows 2014). This biofilm-dispersing capacity of DNaseI to facilitate BAC access into the biofilm is especially relevant in *L. monocytogenes*-carrying biofilms as this bacterium is usually located in the bottom layers (Almeida et al. 2011). In *E. coli*, a better performance of BAC after enzymatic treatment was also observed, but to a lesser extent (Figure 4). This can be attributed to its intrinsic higher resistance to QACs (McDonnell & Russell 1999; Augustin et al. 2004) and also because of the possible presence of protective colanic acid capsules (Miajlovic & Smith 2014).

It is important to remark on the fact that BAC effects against 48 h samples were different depending on the species (Figure 4), whereas in 24 h biofilms DNaseI was the most efficient enzyme in both species of the mixed biofilms (Figure 3). This indicates that the biofilm matrix varies its molecular composition with time. So, if proper enzyme-based biofilm cleaning strategies are intended to be designed it is important to determine the constituents (proteins, eDNA and polysaccharides) of the matrix of the target sessile community.

A release of live cells of both species is observed from biofilms after PRN–BAC or DNaseI–BAC treatments, especially at low BAC concentrations (Figure 6). Pathogen dispersal after sanitation is a factor to take into account in cleaning and disinfection methodologies (Cordeiro & Werner 2011; Nguyen & Burrows 2014). This can be

minimised by using appropriate effective concentrations of disinfectants (eg BAC) after dispersing agents (enzymes in this particular case) to avoid dissemination of live cells in adjacent areas after biofilm removal.

Another important issue to be assessed in dual-species biofilms of *L. monocytogenes* is the role of the accompanying species. Significant differences were observed in the *L. monocytogenes* AVC counts, as well as in the effect of the enzyme and enzyme–BAC treatment, depending on the accompanying bacterium (Figure 6). Regarding the first, a higher number of *L. monocytogenes* A1 cells was attached to SS after 48 h in the presence of *P. fluorescens* relative to *E. coli*, probably due to entrapping of the *L. monocytogenes* into the polymeric matrix secreted by the *P. fluorescens*. Morphological features agreed with previously reported data in which *L. monocytogenes*–*E. coli* biofilms appeared as uniform layers (Almeida et al. 2011) whereas *L. monocytogenes*–*P. fluorescens* were characterised by local microcolony formation surrounded by smaller biofilm aggregates randomly distributed (Figure 7) (Puga et al. 2014).

DNaseI produced a significant decrease of *L. monocytogenes* only in the mixed biofilm with the presence of *E. coli*, probably because the matrix composition affected its diffusion and effectiveness differently (Figure 6) (Allison 2003). Nevertheless, the application of BAC against *L. monocytogenes* was more effective when co-cultured with *P. fluorescens*, despite the latter being considered a strong biofilm former (Figure 6) (Allison et al. 1998; Jackson et al. 2001).

In summary, in this work the effectiveness of treatments with an enzyme solution alone and combined with a BAC dose on *L. monocytogenes* dual-species biofilms was demonstrated. In addition to this, the results demonstrated that the removal efficacy of a combined enzyme–BAC treatment against mixed biofilms depends not only on the enzyme chosen but also on the biofilm species composition. Following this idea, for proper biofilm removal in food related surfaces as well as in others capable of harbouring bacterial biofilms, customised treatments depending on the species composition should be considered when developing new cleaning and disinfection methodologies. This would be intended not only to impede biofilm formation but also to significantly remove structures already present while minimising the amounts of live cells released.

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